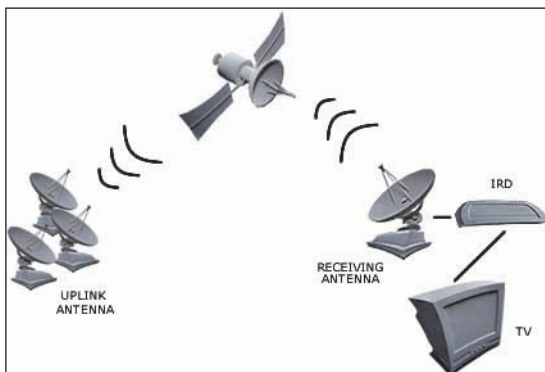


Let us now take a brief look at how satellite broadcasting works. Today, most customers use the direct to home model, where many TV channels are broadcasted by a direct broadcast satellite provider (such as Tata Sky). At the broadcast centre, or the Payout and Uplink location, the various channels are compressed to make the data stream small enough for satellites to handle, encrypted so that non paying receivers cannot receive the signal and transmitted to a satellite in geostationary orbit. A geostationary orbit is one in which the time period for one revolution of a satellite is exactly the same as the rotation of the Earth. Hence the satellite will remain over one point on the earth's surface always, serving as a highly elevated repeating station.

The satellite receives the signals, amplifies them and beams them back onto the earth's surface where dish antennas pick them up and focus them onto the LNB. The LNB is the small box like device found at the centre of dish antennas. Standing for low noise blocker, the function of an LNB is to filter out the noise (non data-carrying signals). The LNB forwards the filtered signal to the set top box which then decrypts the signal using a decryption chip provided by the service provider (a card that can be inserted into a slot on the set top box, in most cases). The signal now decrypted and converted to analogue for playback by a standard TV, is sent to the TV for playback.

The recent rise in popularity of large screens and projectors has rendered many television broadcasting technologies



A schematic diagram of satellite broadcasting.